C6 Semantic models and Direct Lake

Star schemas, DAX best practices and Power BI reports.

C6-L1 Star schemas for semantic models

Facts with a stated grain, dimensions with unique keys, one-to-many single-direction relationships, and derived columns computed upstream.

Star schema parts	Fact tables (keys and measures, with a grain) and dimension tables (unique key, descriptive columns).
Relationship guidance	One-to-many, single direction. Minimise bi-directional relationships.
Calculated columns and Direct Lake	Not supported on SQL endpoints, preview on OneLake. Compute derived columns in gold.
Grain	Set by the fact table's dimension keys. State it in the table's docstring and comment.

C6-L2 DAX measures that state their filter context

Measures, not calculated columns. DIVIDE for every ratio, variables for clarity, and CALCULATE with the narrowest filter change.

```
Weighted aware % =
VAR aware_w = SUMX ( fact_awareness, fact_awareness[aware] * fact_awareness[weight] )
VAR base_w  = SUM ( fact_awareness[weight] )
RETURN DIVIDE ( aware_w, base_w )
```

Measure vs calculated column	Measures compute at query time in the visual's filter context. Calculated columns are stored at refresh.
DIVIDE in DAX	Returns BLANK (or an alternate result) when the denominator is zero or BLANK.
VAR ... RETURN	Evaluates once, names parts, improves performance, readability and debugging.
Share of total across brands	CALCULATE([measure], ALL(dim_brand)): remove only the brand filter.

C6-L3 Direct Lake, framing and the fallback you did not plan

Direct Lake stays fast when tables are Delta (not views), compact, within the SKU's guardrails, and fallback is made visible in development.

Direct Lake refresh	Framing: records Delta table versions in seconds. No data copy.
Direct Lake on OneLake vs on SQL	OneLake: never falls back. SQL: falls back to DirectQuery for views and SQL-based security.
Direct Lake behavior values	Automatic (default, silent fallback), DirectLakeOnly (error), DirectQueryOnly (always DirectQuery).
F64 per-table guardrails	5,000 Parquet files, 1.5 billion rows, 25 GB max memory.